

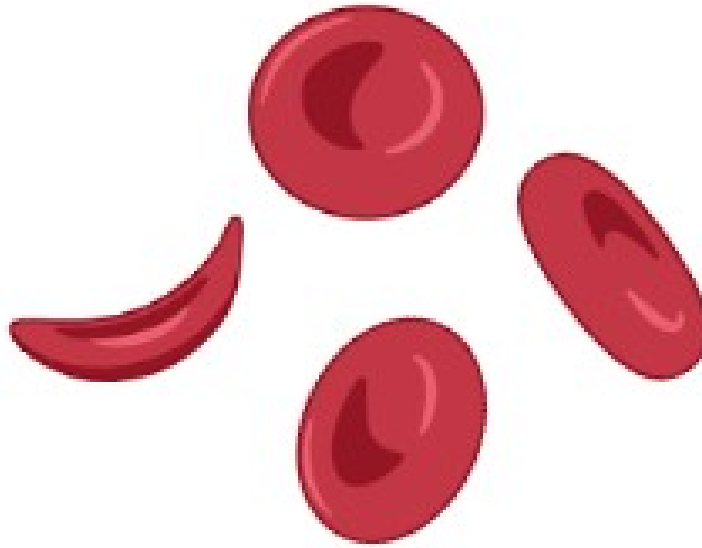
Overview

Underprod

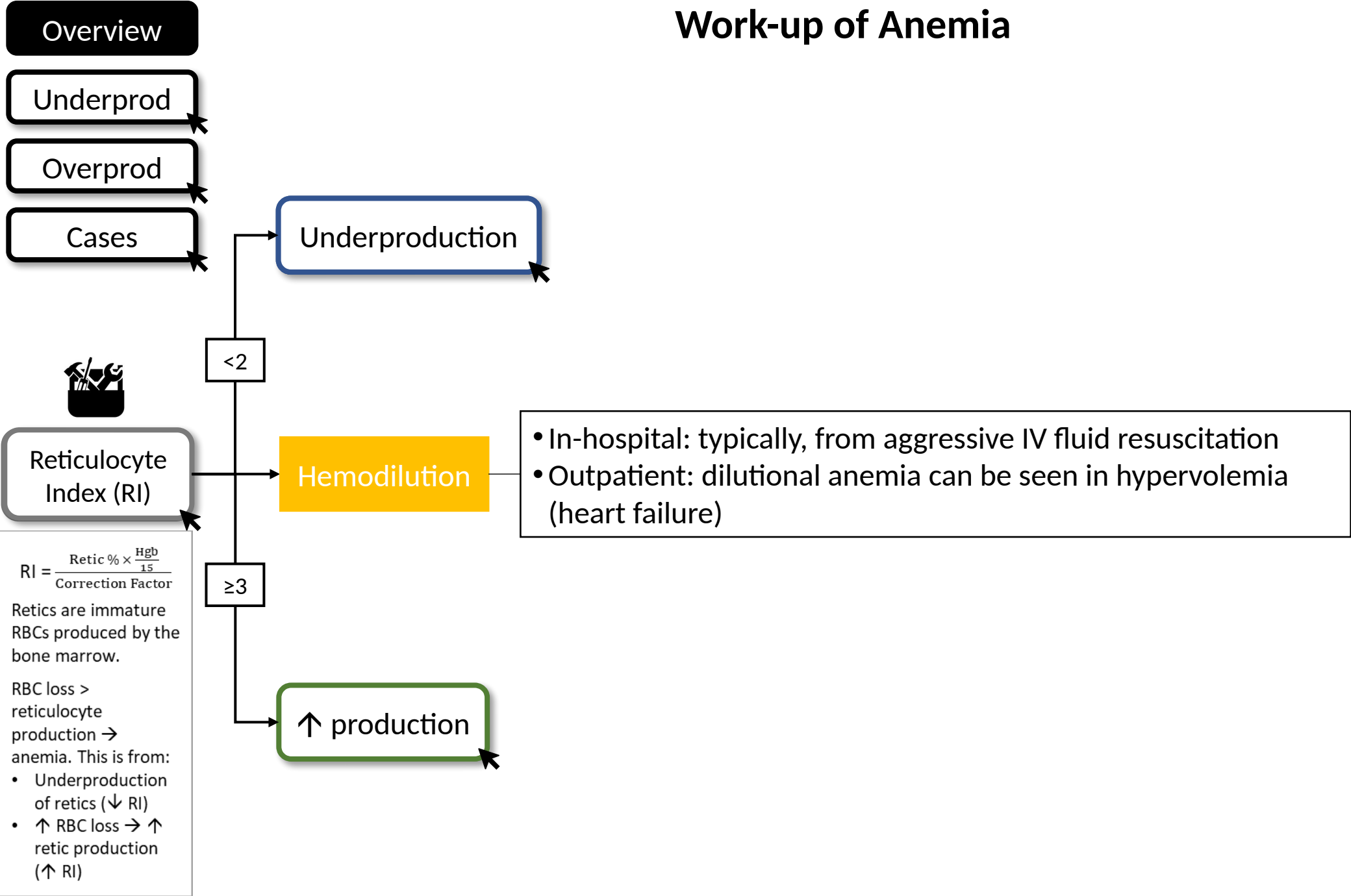
Overprod

Cases

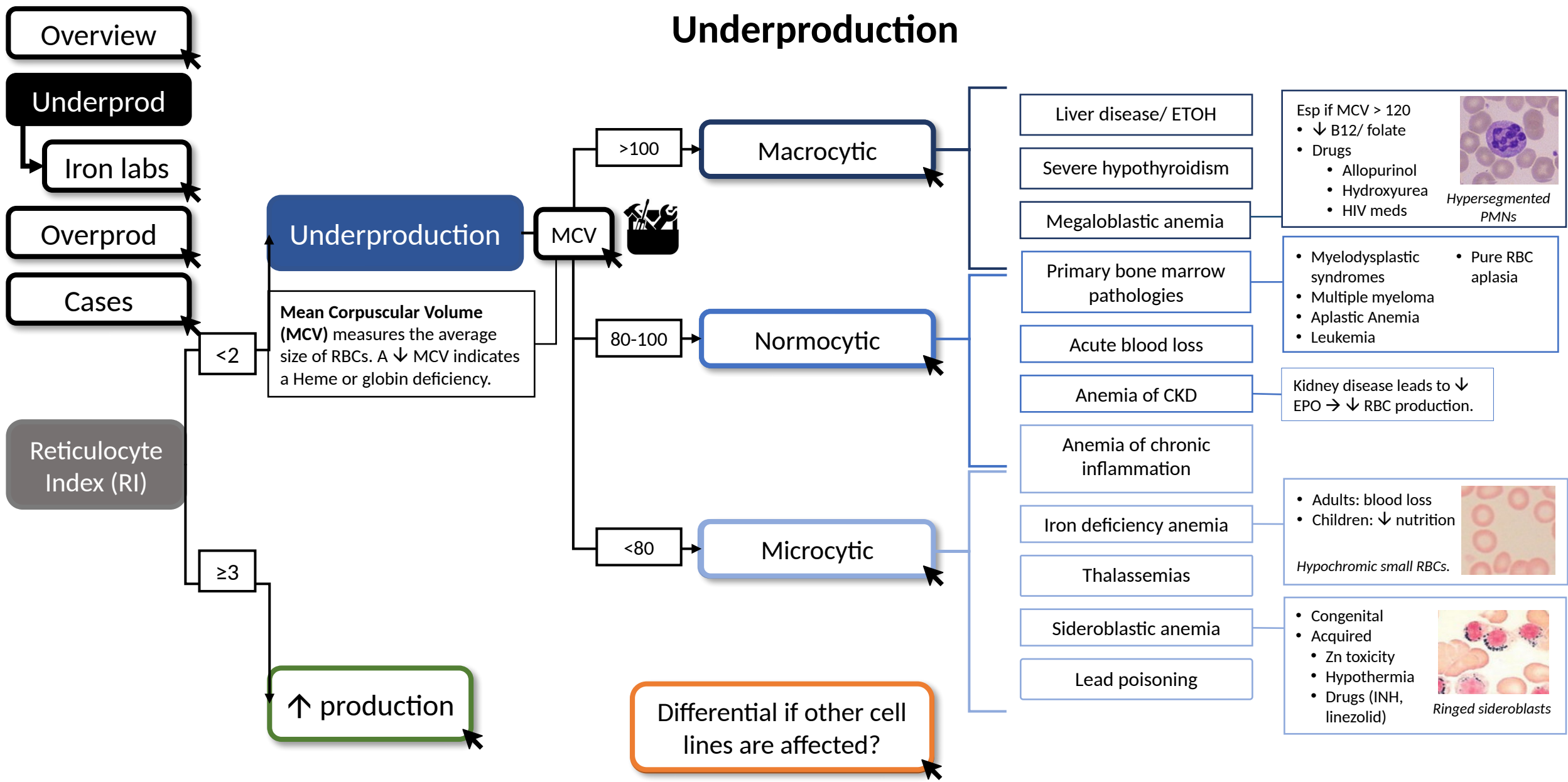
Work-up of Anemia



Work-up of Anemia



Underproduction



Overview

Underprod

Iron labs

Overprod

Cases

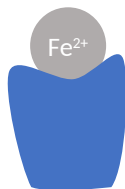
Underproduction – Iron Studies



Iron studies can help differentiate between microcytic causes.

Serum Iron

- Levels fluctuate with dietary intake
- Fasting level most reliable



TIBC

Correlates to transferrin levels (circulating transport protein for iron)

% saturation

- Serum iron $\div$ TIBC
- Normal: 25-45%



Ferritin

- Measure of iron stores (correlate of bone marrow iron)
- Acute phase reactant

Iron deficiency



(may vary with recent oral intake)



(in response to low iron stores)



(% sat <20% suggestive, <10% very suggestive)



(<30 mg/L is diagnostic)

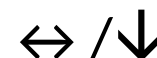
ACD



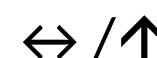
(↓ dietary absorption and sequestration in macrophages)



(inflammatory cytokines ↓ transferrin)



(% sat <20% in 80% of cases but may be pseudo-normal if ↓ TIBC)



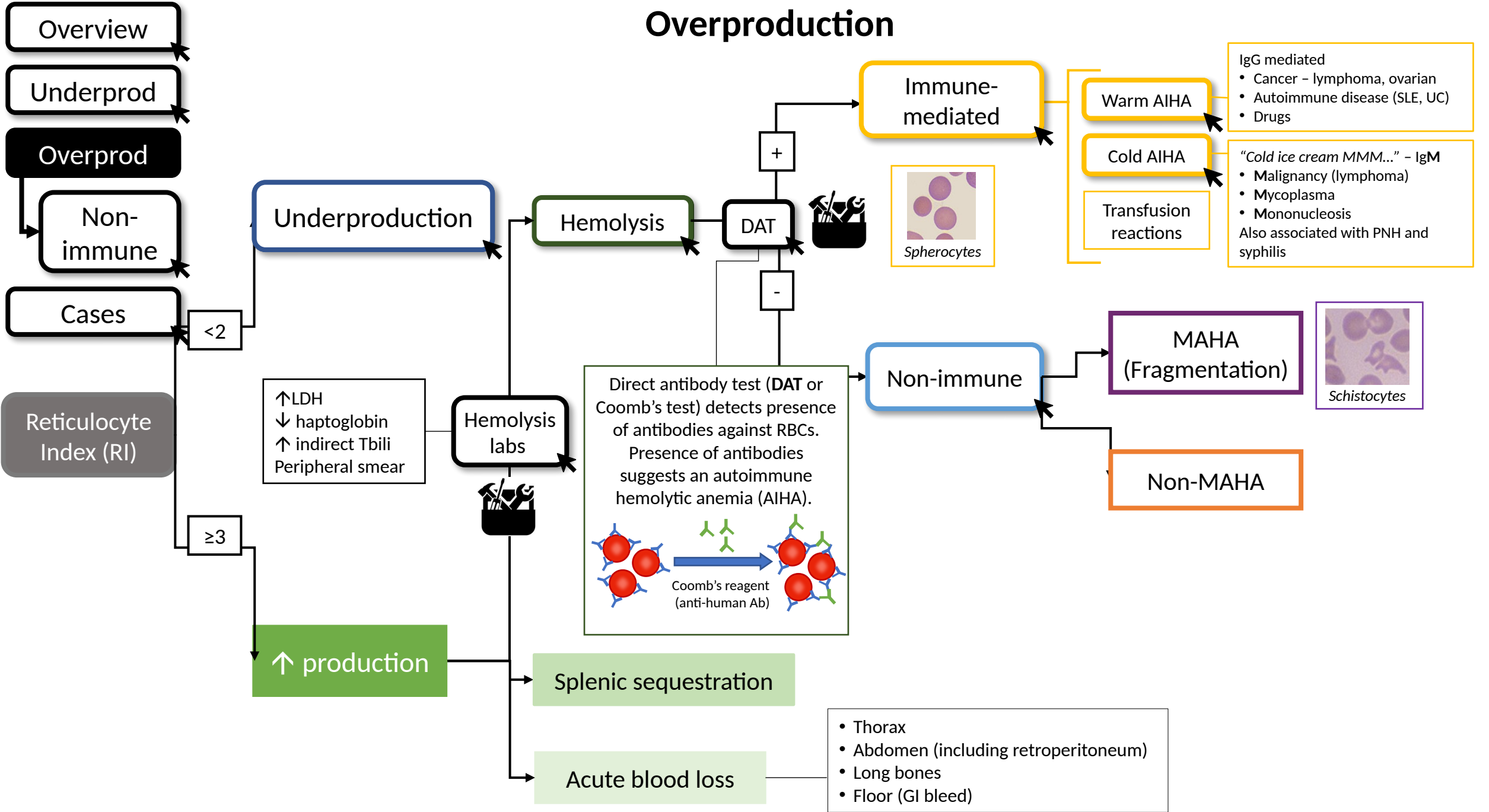
(normal stores, ↑ secretion by macrophages, and inflammatory state)

What if there is concomitant iron deficiency and ACD?

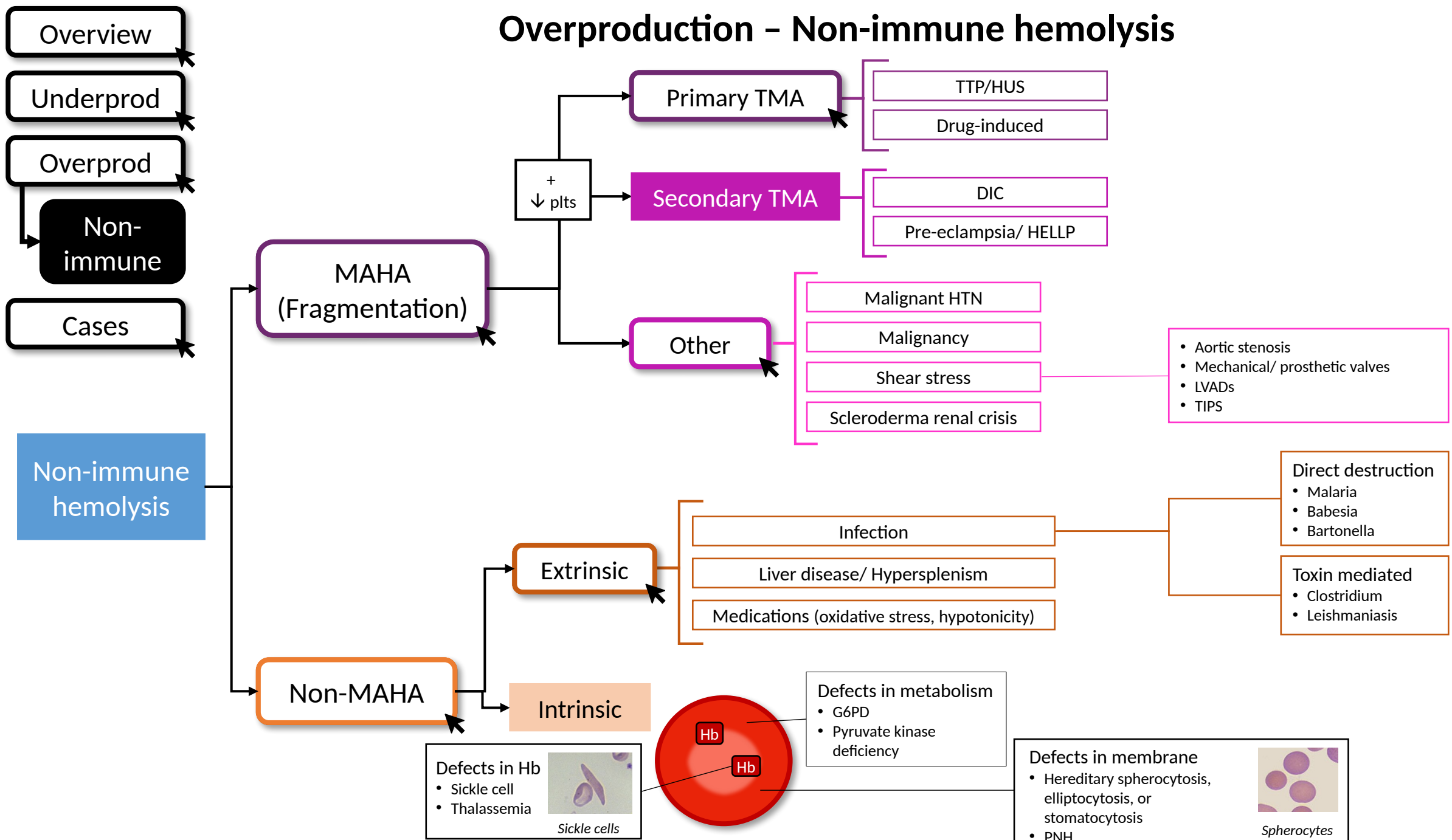
20-85% of patients with ACD also have iron deficiency.

Hepcidin, soluble transferrin receptor (sTfR) or response to trial of iron can be used to differentiate.

Overproduction



Overproduction - Non-immune hemolysis



Overview

Underprod

Overprod

Cases

1-2

3-4

Case 1

56-year-old presents with a ground level fall and femur fracture.

ROS: No bleeding. Reports weakness, fatigue, mild confusion, constipation, and easy bruising

Cr 1.6
Calcium 11.5
Retic % 4.5

3.9 8.2 147
25
27 15
1.1



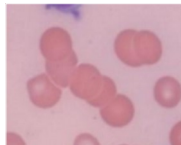
RI = 1.1

MCV = 89

Serum Fe ↓, % sat ↓, ferritin ↑

LDH 1508, haptoglobin nml, Tbili 1.0 (direct 0.3)

Not sent.



Rouleaux formation

Multiple myeloma (hyperCa, Renal dysfunction, Anemia, Boney lesions). Iron studies show anemia of chronic disease. LDH is ↑ from cell turnover.

Case 2

33-year-old presents with 2 weeks of fevers/ chills, nausea/ vomiting and non-bloody diarrhea.

ROS: + fatigue, + menorrhagia, + gum bleeding.

Exam: T 36 C, HR 90, BP 123/54, blood blister on lips.

Cr 1.0
Fibrinogen nml

7.9 5.1 9
13
43 16.1
1.3



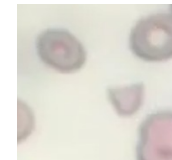
RI = 3.5

MCV = 89

Serum Fe ↓, % sat ↓, ferritin ↓

LDH 1508, haptoglobin < 30, Tbili 1.0 (direct 0.3)

DAT negative.



Schistocytes. Suggestive of a MAHA.

Primary TMA (e.g., TTP) given MAHA + ↓ platelets.
This is a hematologic emergency!
There is concurrent iron deficiency anemia.

Overview

Underprod

Overprod

Cases

1-2

3-4

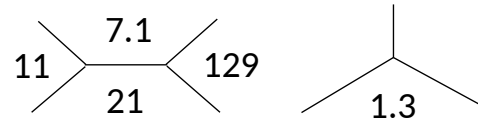
Case 3

45-year-old with SLE presents with chest pain and shortness of breath.

ROS: + jaundice, joint pains, fatigue

Exam: T 36C, HR 90, BP 123/54, lungs clear, no rubs/ murmurs.

Cr 1.6



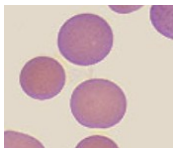
RI = 4.5

MCV = 115

Serum Fe ↓, sat ↓, ferritin ↑

LDH 2504, haptoglobin < 30, Tbili 4.6 (direct 0.3)

DAT positive. IgG positive.



Spherocytes.

Warm AIHA associated with SLE. ↑ MCV may be from reticulocytosis. Iron studies show concurrent anemia of chronic disease.

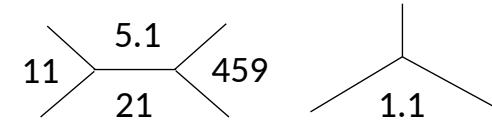
Case 4

73-year-old with minimal health care contact presents with 6 months of progressive fatigue and dyspnea on exertion.

ROS: normal diet, no overt blood loss.

Exam: T 36C, HR 90, BP 123/54, SaO2 100% on room air. Lungs clear, no rubs/ murmurs.

Cr 1.0



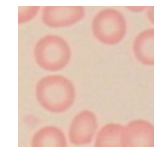
RI = 1.1

MCV = 68

Serum Fe ↓, sat 1%, TIBC normal, ferritin 150 mg/L (↑)

LDH 254, haptoglobin normal, Tbili 1.0 (direct 0.3)

Not sent.



Hypochromic RBCs.

Concurrent iron deficiency (very low % sat) and **anemia of chronic disease** (normal TIBC, ↑ ferritin). Further GI evaluation for occult blood loss is needed.

Framework Update & High-Yield Refinements (2019–25)

Iron studies — refine the cutoffs:

Ferritin <30 ng/mL = iron deficiency (highly specific). In inflammation/CKD use a higher cutoff (<100, or TSAT <20%); soluble transferrin receptor or sTfR/log-ferritin index and hepcidin help when ACD + iron deficiency coexist (20–85% of ACD).

IV iron (ferric carboxymaltose, ferric derisomaltose) preferred over oral when malabsorption, ongoing loss, CKD/HF, or oral intolerance — faster repletion (watch hypophosphatemia with FCM).

MAHA + thrombocytopenia = possible TMA — act fast:

Use the PLASMIC score to gauge TTP probability and send ADAMTS13 BEFORE plasma exchange; do not wait for the level if TTP is likely (caplacizumab + PEX + steroids is now standard for confirmed TTP).

Schistocytes + low platelets + normal coags → primary TMA (TTP/HUS); add fibrinogen/D-dimer to separate DIC; check pregnancy/HELLP and complement-mediated (atypical HUS).

Diagnostic framework otherwise unchanged: retic index → under- vs over-production; MCV + iron studies for underproduction; DAT + smear for hemolysis.

Take-Home Points

Start every anemia work-up with the reticulocyte index: <2 = underproduction (marrow problem), ≥ 3 = overproduction (blood loss, hemolysis, or sequestration).

For underproduction, use MCV (macro/normo/microcytic) plus iron studies; ferritin <30 ng/mL is diagnostic of iron deficiency but is an acute-phase reactant — interpret it against inflammation.

For overproduction, confirm hemolysis ($\uparrow$ LDH, $\downarrow$ haptoglobin, $\uparrow$ indirect bili) then use the DAT + smear: DAT-positive = autoimmune (warm IgG / cold IgM); schistocytes = MAHA.

MAHA with thrombocytopenia is a hematologic emergency — think TMA (TTP/HUS), send ADAMTS13, and don't delay plasma exchange when TTP is likely.

Concurrent iron deficiency + anemia of chronic disease is common (a very low TSAT is the clue) — a trial of iron with repeat labs often settles it.

Attribution & Optimization Note

Adapted from TeachIM.org — “Work-up of Anemia” (Jeff Krimmel-Morrison MD, Sioban Keel MD; exec. eds. Yilin Zhang MD, Brandon Fainstad MD). Published Mar 2020; deck rev. Dec 2022.

Source: teachim.org/teaching_material/work-up-of-anemia-chalk-talk-teachim/ · License: CC BY-NC 4.0 (attribution, non-commercial only).

Correction made (this copy): slide 6 — “stomacytosis” → “stomatocytosis” (RBC-membrane defect). All original animations/click-to-reveal builds and graphics preserved; no other slide content altered.

Added context (source-cited): inflammation-adjusted ferritin/TSAT cutoffs + sTfR/hepcidin (Weiss, Blood 2019; Camaschella, Blood 2019); IV iron preference; PLASMIC score + ADAMTS13-before-PEX and caplacizumab for TTP.

Companion file: Anemia-Handout-12.22.pdf (learner handout + fill-in schematic). Educational use only; not medical advice — have content clinician-reviewed before teaching.